

Microsoft Entra IAM Architecture Project

Project Author

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Environment

Microsoft Entra ID (Free Tier)

Focus

Identity Architecture, RBAC Design, Identity Lifecycle Governance, and Zero Trust Conditional Access Planning.

Project Overview

This project demonstrates the design and implementation of a secure Identity and Access Management (IAM) architecture using Microsoft Entra. The goal of the project was to design an enterprise-style identity architecture that follows Zero Trust principles while operating within the constraints of a free tenant environment.

The project focuses on architecture design rather than simple configuration tasks. It demonstrates how identity systems should be structured in real organizations, including role separation, lifecycle management, governance monitoring, and Conditional Access policy planning.

Although the environment operates under Microsoft Entra ID Free licensing, the architecture documentation reflects a production-ready model that would normally operate with Entra ID P1/P2 licensing.

Objectives

The primary objectives of this project were to design a scalable IAM architecture using Microsoft Entra. Enforce role-based access control (RBAC). Separate privileged and non-privileged identities. Implement group-based identity organization. Simulate joiner, mover, and leaver identity lifecycle events. Demonstrate governance through audit logging and to design Conditional Access policies aligned with Zero Trust security principles.

Identity Architecture Design

The IAM architecture was designed to follow modern identity governance practices used in enterprise environments.

Identity Boundary

All workforce identities are managed within a centralized Microsoft Entra tenant.

Access to administrative roles, enterprise applications, and organizational resources is governed through identity controls, role assignments, and audit monitoring.

Administrative Tier Model

A simplified administrative tier model was implemented to reduce the risk of privilege misuse.

Tier 0 — Privileged Administration

Highly restricted identities responsible for tenant-level administration.

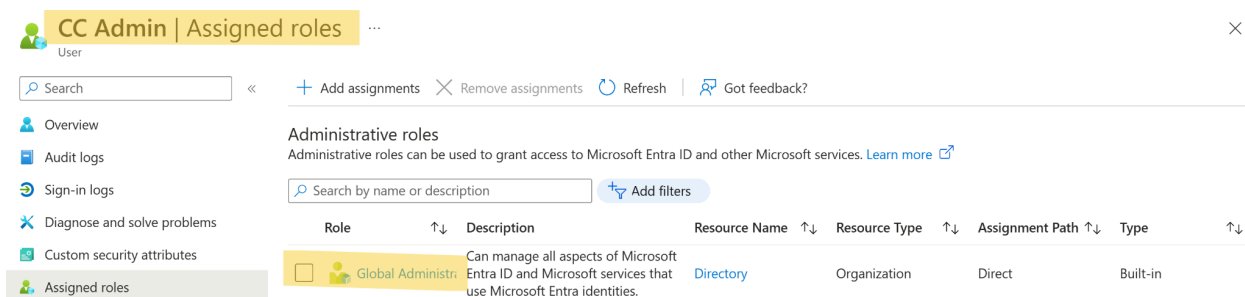
Example:
cc-admin

Responsibilities include:

- Global Administrator access
- Tenant configuration

- Identity governance oversight

Figure 1: Global Administrator role assigned to privileged Tier 0 account.



The screenshot shows the 'Assigned roles' page for the user 'CC Admin'. The left sidebar lists navigation options: Overview, Audit logs, Sign-in logs, Diagnose and solve problems, Custom security attributes, and Assigned roles. The main content area is titled 'Administrative roles' and includes a search bar and a table of assigned roles.

Role	Description	Resource Name	Resource Type	Assignment Path	Type
Global Administrator	Can manage all aspects of Microsoft Entra ID and Microsoft services that use Microsoft Entra identities.	Directory	Organization	Direct	Built-in

Tier 1 — IT Operations

Operational administrative identities responsible for user management and support.

Examples:

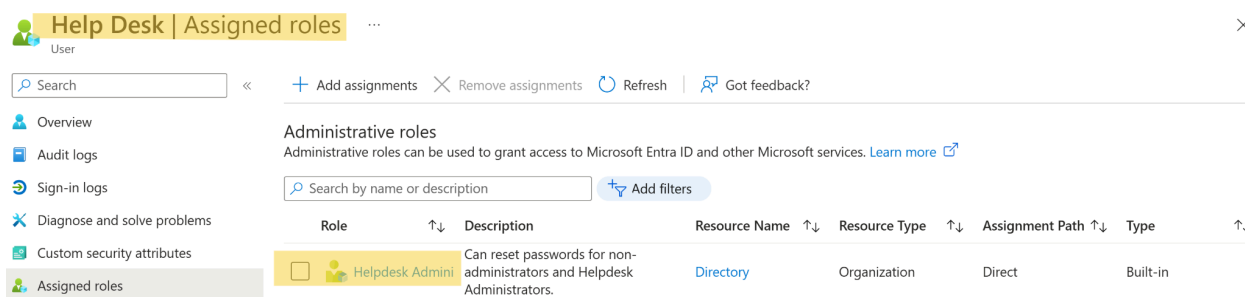
helpdesk1

itops1

Responsibilities include:

- User account management
- Password resets
- Identity troubleshooting

Figures 2 & 3: Tier 1 operational roles assigned to helpdesk and IT operations accounts.



The screenshot shows the 'Assigned roles' page for the user 'Help Desk'. The left sidebar lists navigation options: Overview, Audit logs, Sign-in logs, Diagnose and solve problems, Custom security attributes, and Assigned roles. The main content area is titled 'Administrative roles' and includes a search bar and a table of assigned roles.

Role	Description	Resource Name	Resource Type	Assignment Path	Type
Helpdesk Administrator	Can reset passwords for non-administrators and Helpdesk Administrators.	Directory	Organization	Direct	Built-in

ITops | Assigned roles

User

Search

Add assignments
Remove assignments
Refresh
Got feedback?

Overview

Audit logs

Sign-in logs

Diagnose and solve problems

Custom security attributes

Assigned roles

Administrative roles

Administrative roles can be used to grant access to Microsoft Entra ID and other Microsoft services. [Learn more](#)

Search by name or description

Add filters

Role	Description	Resource Name	Resource Type	Assignment Path	Type
User Administrator	Can manage all aspects of users and groups, including resetting passwords for limited admins.	Directory	Organization	Direct	Built-in

Tier 2 — Standard Users

Regular workforce identities with no administrative privileges.

Examples:

- exec1
- contractor1
- cc-user

Figures 4, 5, 6: Standard workforce identities with no administrative privileges.

Users

Default Directory

New user

Edit

Delete

Download users (Preview)

Bulk operations

Refresh

Manage view

Azure Active Directory is now Microsoft Entra ID.

Search

Add filter

7 users found

Display name	User principal name	User type	Is Agent	On-premises sy...	Identities
CA CC Admin	cc-admin@...	Member	No	No	
CU CC User	cc-user@...	Member	No	No	
C Contractor	contractor@...	Member	No	No	
E Exec	exec@...	Member	No	No	
HD Help Desk	helpdesk@...	Member	No	No	
I ITops	itops@...	Member	No	No	

Role-Based Access Control (RBAC)

RBAC was implemented to enforce least-privilege access.

Due to Microsoft Entra Free licensing, directory roles were assigned directly to users. In a licensed production environment, roles would instead be assigned through role-based security groups.

Implemented Role Assignments

Helpdesk Administrator → helpdesk1

User Administrator → itops1

Global Administrator → cc-admin

This structure demonstrates separation of duties and reduces unnecessary privilege exposure.

Figures 7 & 8: Role-based access control implementation within Microsoft Entra.

Help Desk | Assigned roles

User

Search

«

+ Add assignments

✕ Remove assignments

🔄 Refresh

🗨 Got feedback?

Overview

Audit logs

Sign-in logs

Diagnose and solve problems

Custom security attributes

Assigned roles

Administrative roles

Administrative roles can be used to grant access to Microsoft Entra ID and other Microsoft services. [Learn more](#)

Search by name or description

Add filters

Role	↑↓	Description	Resource Name	↑↓	Resource Type	↑↓	Assignment Path	↑↓	Type	↑↓
Helpdesk Admin		Can reset passwords for non-administrators and Helpdesk Administrators.	Directory		Organization		Direct		Built-in	

ITops | Assigned roles

User

Search

«

+ Add assignments

✕ Remove assignments

🔄 Refresh

🗨 Got feedback?

Overview

Audit logs

Sign-in logs

Diagnose and solve problems

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Role	↑↓	Description	Resource Name	↑↓	Resource Type	↑↓	Assignment Path	↑↓	Type	↑↓
User Administrator		Can manage all aspects of users and groups, including resetting passwords for limited admins.	Directory		Organization		Direct		Built-in	

Identity Group Architecture

Security groups were created to represent organizational identity populations and future Conditional Access scopes.

Population Groups

GRP-POP-Employees

GRP-POP-Contractors

These groups represent workforce identity types.

Figures 9 & 10: Workforce population groups created to categorize employee identity types.

The screenshot displays the 'GRP-POP-Contractors' group in the Microsoft Entra ID portal. The left-hand navigation pane includes sections for 'Overview' (selected), 'Diagnose and solve problems', and 'Manage' (containing Properties, Members, Owners, Roles and administrators, Administrative units, Group memberships, and Applications). The main content area shows the group's name with a red 'G' icon, a 'Delete' button, and a 'Got feedback?' link. Below this, a table lists the group's properties:

Membership type	Assigned	Total direct members	1
Source	Cloud	User(s)	1
Type	Security	Group(s)	0
Object ID	4f6ec95a-de62-4bcb-8dc7-a2e52f1cb48b	Device(s)	0
Created on	3/3/2026, 10:29 AM	Other(s)	0

The screenshot displays the 'GRP-POP-Employees' group in the Microsoft Entra ID portal. The left-hand navigation pane includes sections for 'Overview' (selected), 'Diagnose and solve problems', and 'Manage' (containing Properties, Members, Owners, Roles and administrators, Administrative units, Group memberships, Applications, and Licenses). The main content area shows the group's name with an orange 'G' icon, a 'Delete' button, and a 'Got feedback?' link. Below this, the 'Overview' tab is active, showing 'Basic information' and a table of properties:

Membership type	Assigned	Total direct members	4
Source	Cloud	User(s)	4
Type	Security	Group(s)	0

Conditional Access Scope Groups

GRP-CA-Admins

GRP-CA-Users

GRP-CA-Contractors

GRP-EXCLUDE-BreakGlass

These groups define how Conditional Access policies would be scoped in a licensed environment.

Figure 11: Conditional Access policy scope groups defined for future policy enforcement.

☐	 GRP-CA-Admins	244888e7-ea8e-45dc-8e76-0977dc803672	Security	Assigned
☐	 GRP-CA-Contractors	5cca4223-5033-4544-8468-67aafa73e7c	Security	Assigned
☐	 GRP-CA-Users	985896a9-ecfa-450d-8612-61fc61b2c063	Security	Assigned
☐	 GRP-EXCLUDE-BreakGlass	5c64e27c-c9d0-4991-9977-60db10084c8c	Security	Assigned

Enterprise Application Access Model

A simulated enterprise application (Demo-SaaS-App) was created to demonstrate application access management.

In the lab environment, application access was assigned directly to users due to licensing limitations.

Production Design Model

Users → Application Access Groups → Enterprise Applications

This model allows scalable provisioning, simplifies lifecycle management, and improves audit visibility.

Figure 12: Simulated enterprise application used to demonstrate identity-based access control.

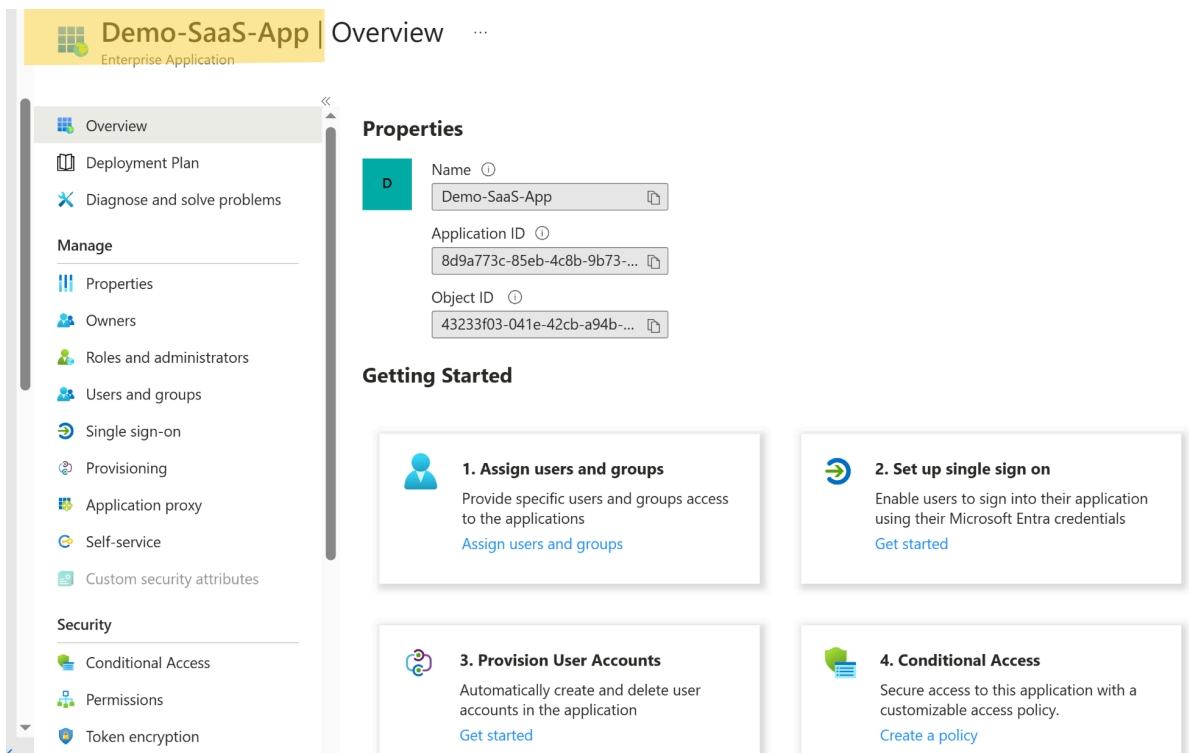
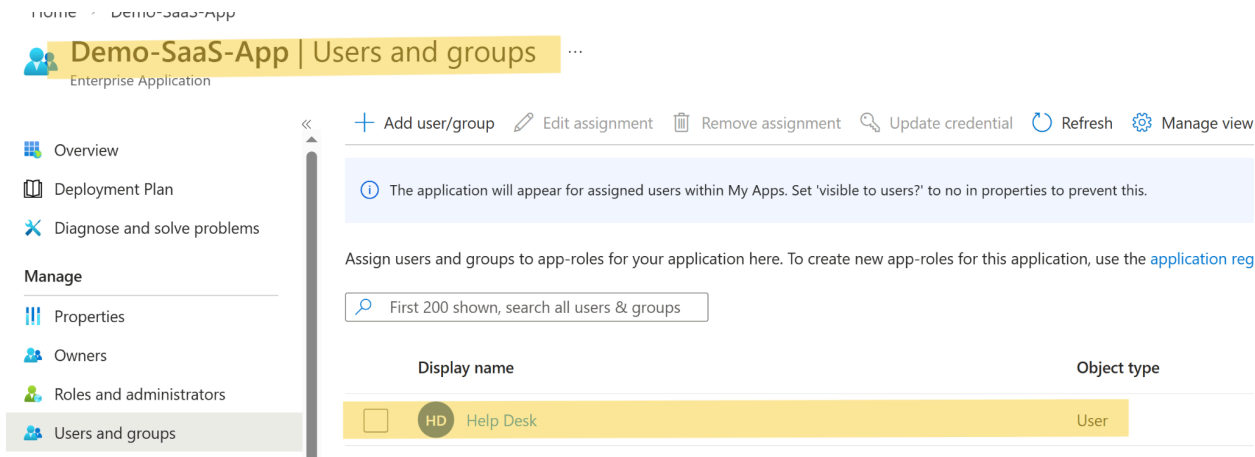


Figure 13: Application access assigned to a user within the lab environment.



Identity Lifecycle Management

Identity lifecycle scenarios were simulated to demonstrate joiner, mover, and leaver workflows.

Joiner Scenario

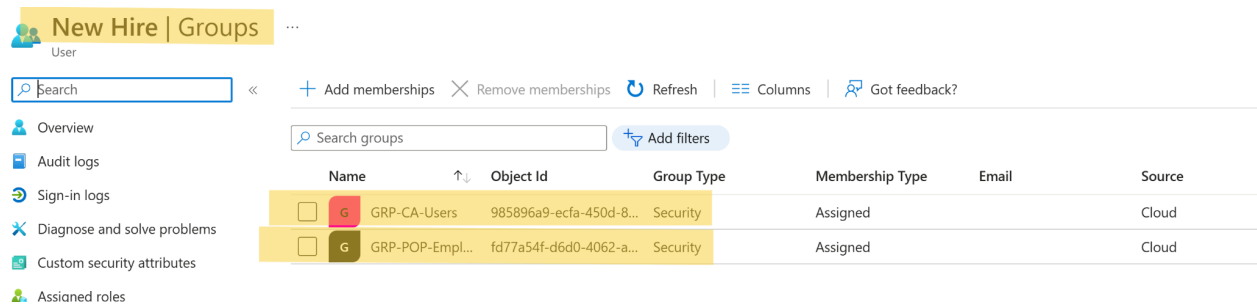
A new user account was created to simulate employee onboarding.

Provisioning included:

- User account creation
- Population group assignment
- Application access assignment

This demonstrates how new identities inherit permissions based on group membership.

Figure 14: Newly provisioned workforce identity created during joiner simulation.



Mover Scenario

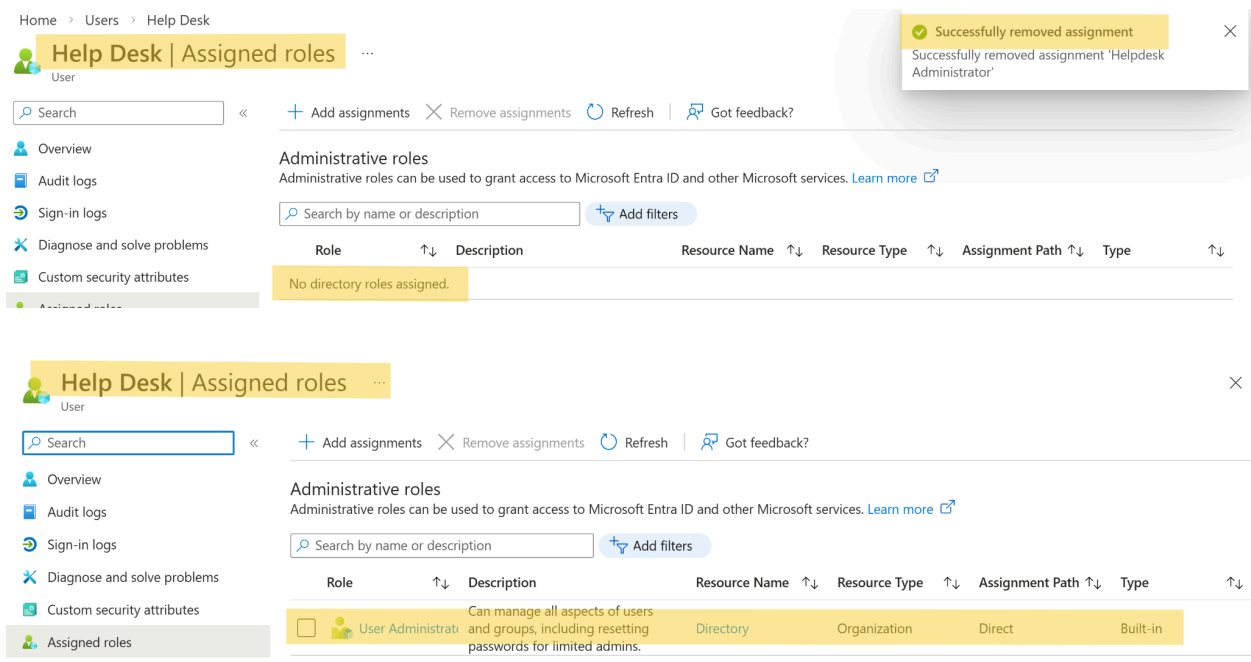
A role change was simulated by modifying administrative permissions.

Example:

helpdesk1 transitioned to IT operations responsibilities.

Role reassignment demonstrates how access changes can be managed through identity governance processes.

Figure 15 & 16: Administrative role reassignment demonstrating identity role transition.



Leaver Scenario

A termination scenario was simulated for contractor1.

The offboarding process followed secure identity governance practices.

Steps performed:

1. Sign-in access disabled (Account Enabled set to No)
2. Application access removed
3. Group memberships removed
4. Access changes verified in audit logs

This workflow demonstrates secure identity offboarding and privilege revocation.

Figure 17: Disabled account used to simulate secure identity offboarding.

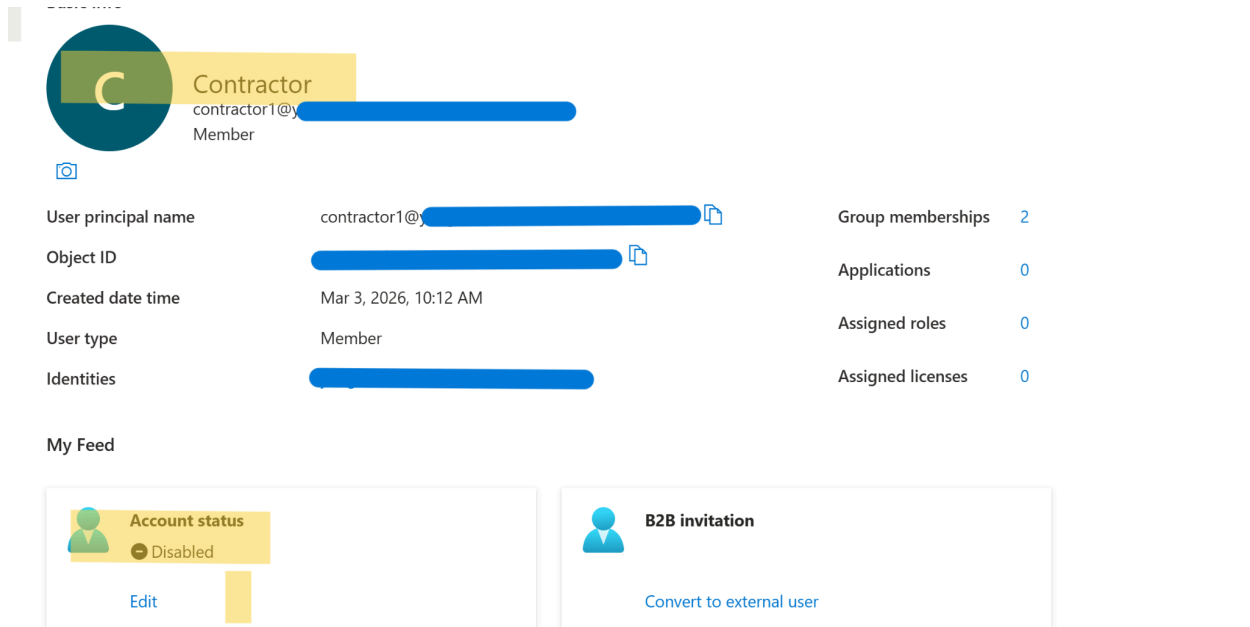


Figure 18: Audit log entry capturing account disablement event.

The screenshot shows the 'Audit Logs' page in Microsoft Entra. The page has a navigation bar with 'Home', 'Users', 'Contractor', 'Audit logs', and 'Microsoft Entra'. The 'Audit Logs' section is active. Below the navigation bar, there are filters for 'Download', 'Export Data Settings', 'Refresh', 'Manage view', and 'Got feedback?'. A message states: 'Want to switch back to the legacy audit logs experience? Click here to leave the preview.' Below this, there are filter buttons: 'Add filter', 'Show dates as: Local', 'Date range: Last 24 hours', 'Service: All', 'Category: All', 'Activity: All', 'Target starts with contractor1', and 'Reset filters'. The table below shows a list of audit events.

Date ↓	Service	Category	Activity	Status	Status Reason
3/3/26, 3:20:10 PM	Core Directory	UserManagement	Update user	Success	
3/3/26, 3:20:10 PM	Core Directory	UserManagement	Disable account	Success	
3/3/26, 10:45:08 AM	Core Directory	GroupManagement	Add member to group	Success	
3/3/26, 10:45:08 AM	Core Directory	GroupManagement	Add member to group	Success	
3/3/26, 10:12:24 AM	Core Directory	UserManagement	Add user	Success	
3/3/26, 10:12:24 AM	Core Directory	UserManagement	Update PasswordProfile	Success	

9. Governance and Auditability

Identity governance was validated through Microsoft Entra audit logs.

Administrative actions captured included:

- User account modifications
- Role assignments
- Group membership changes
- Identity lifecycle actions

Each audit record includes:

Actor

Action

Target

Timestamp

This provides traceability and accountability for all identity changes.

Figure 14: Identity governance evidence showing administrative action recorded in audit logs.

Audit Logs ...

[Download](#) [Export Data Settings](#) [Refresh](#) [Manage view](#) [Got feedback?](#)

Want to switch back to the legacy audit logs experience? [Click here to leave the preview.](#)

[Add filter](#) [Show dates as: Local](#) [Date range: Last 24 hours](#) [Service : All](#) [Category : All](#) [Activity : All](#) [Reset filters](#)

Directory

Custom Security

Date ↓	Service	Category	Activity	Status	Status Reason
3/3/26, 6:16:51 PM	Self-service Group Man...	GroupManagement	Group_GetDynamicGrou...	Success	OK
3/3/26, 6:11:17 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK
3/3/26, 6:11:17 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK
3/3/26, 6:11:16 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK
3/3/26, 6:11:14 PM	Self-service Group Man...	GroupManagement	Group_GetDynamicGrou...	Success	OK
3/3/26, 6:07:30 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK
3/3/26, 6:07:30 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK
3/3/26, 6:07:29 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK
3/3/26, 6:07:10 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK
3/3/26, 6:06:42 PM	Self-service Group Man...	GroupManagement	GroupsODataV4_Get	Success	OK

Conditional Access Architecture

Conditional Access policies were designed as part of a security architecture portfolio.

Because the tenant operates under Entra Free licensing, these policies were documented rather than enforced.

Policy 1 — Block Legacy Authentication

Scope: All users except break-glass accounts

Control: Block legacy authentication protocols

Threat Mitigated:

Credential replay attacks and bypass of MFA controls.

Licensing Requirement:

Microsoft Entra ID P1

Policy 2 — Require MFA for Administrators

Scope: Administrative identities (GRP-CA-Admins)

Control:

Require multi-factor authentication for privileged accounts.

Threat Mitigated:

Administrative credential compromise.

Licensing Requirement:

Microsoft Entra ID P1

Policy 3 — Require MFA for Contractors

Scope: GRP-CA-Contractors

Control:

Require MFA during authentication.

Threat Mitigated:

External identity compromise from unmanaged devices or networks.

Licensing Requirement:

Microsoft Entra ID P1

Zero Trust Alignment

The IAM architecture aligns with Zero Trust identity principles.

Key controls include:

- Privileged identity separation
- Strong authentication design (MFA)
- Reduced legacy authentication risk
- Identity lifecycle governance
- Audit logging and accountability

The architecture assumes no identity is inherently trusted and enforces verification through identity controls.

Lab Environment Limitations

The environment used Microsoft Entra ID Free licensing, which restricts certain features.

Limitations included:

- Conditional Access enforcement
- Group-based role assignment
- Group-based application assignment
- Privileged Identity Management (PIM)

These controls were therefore documented as architectural design elements rather than enforced configurations.

Key Security Takeaways

This project demonstrates the importance of:

- Separating administrative identities
- Implementing least-privilege access
- Managing identity lifecycle events securely
- Monitoring identity activity through audit logs
- Designing Conditional Access policies aligned with Zero Trust

The project reflects how identity architecture must balance security controls, operational governance, and platform licensing constraints.

Skills Demonstrated

Microsoft Entra Identity Administration
Identity Architecture Design
Role-Based Access Control (RBAC)
Identity Lifecycle Governance
Conditional Access Design
Zero Trust Security Principles
IAM Audit Monitoring

Project Outcome

The result of this project is a structured IAM architecture demonstrating secure identity management practices within Microsoft Entra.

The project highlights both the technical configuration of identity systems and the architectural planning required to secure enterprise identity infrastructure.